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Parametric Study of Die-Sinking EDM for Titanium Diboride with Varied Electrodes

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Titanium diboride (TiB_2) is a type of ultra-high temperature ceramic (UHTC) known for its exceptional hardness, high melting point, and excellent oxidation resistance. It is ideal for demanding industrial applications like cutting tools, wear-resistant components, and thermal protection in high-temperature manufacturing environments. TiB_2's exceptional hardness and brittleness pose problems for traditional machining processes, resulting in excessive tool wear and potential workpiece damage. Electrical Discharge Machining (EDM) die sinking is a potential option for precise shape of electrically conductive ceramics without producing mechanical stress. This study analyzes die-sinking EDM of TiB_2 using copper, graphite, and copper-tungsten electrodes. A Taguchi orthogonal array design was used to assess the effect of major process parameters—pulse on time, pulse off time, discharge current, gap voltage, and auxiliary jet flushing height—on material removal rate (MRR), surface roughness (Ra), tool wear rate (TWR), and dimensional overcut. Experiments were conducted on sintered TiB_2 specimens with IONOPLUS 3000 dielectric oil to maintain steady discharge conditions and minimize fume formation.

The findings offer statistical insight into the relationship between electrode material qualities and machining responses, as well as optimal parameter sets for roughing, semi-finishing, and finishing processes. This work provides practical suggestions for efficient EDM processing of TiB_2, making it more suitable for high-performance industrial applications that need precision, surface integrity, and productivity.